National University of Computer and Emerging Sciences
Lahore Campus

Analogue & Digital Communication (EE3003)

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 40
Total Questions: 2

Date: November 03, 2025

Course Instructor(s)

1. Mohsin Yousuf (Course Moderator)
2. Aroosa Umair

Roll No ____________ Section ____________ Student Signature ____________

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1. **Attempt all questions and remember to solve parts of the same question together.**
2. **Show all the steps with the help of diagrams and equations.**

CLO # 02: Apply** the **concept of sampling, analog amplitude** and frequency **modulation and demodulation techniques.

Q1: **[20 marks]**

(a) A modulating signal $m(t)$ is given by, **(10)**

A m ωt

$$m(t) = 10\cos(1500\,t)$$

The carrier angular frequency, ω_c is 10,000 rad/s.

(i) **Determine** the Lower Sideband and Upper Sideband modulated signals denoted by $\varphi_{LSB}(t)$, $\varphi_{USB}(t)$ respectively.

(ii) **Represent** these signals in the form of Single Sideband $\varphi_{SSB}(t)$.

(iii) **Sketch** the corresponding frequency spectrum.

(iv) Mathematically **exposit** that if carrier amplitude, $A \gg m(t)$, $m(t)$ can be recovered from φ_{SSB+C} using envelope or rectifier detection. Also **name** this type of demodulation.

(b) **(i)** **Describe** the process of Sampling and **state** the Sampling Theorem precisely with the importance of Nyquist criterion. **(10)**

(ii) **Demonstrate** ideal interpolation for a signal $g(t)$ band-limited to B Hz both in time and frequency-domain with the help of equations and diagrams.

Hint: $H(\omega) = T_s \operatorname{rect}\left(\frac{\omega}{4\pi B}\right)$

CLO # 02: Apply** the concept of sampling, analog amplitude and **frequency modulation and demodulation techniques

Q2: **[20 marks]**

(a) The modulating signal $m(t)$ is a periodic sawtooth signal as shown in Figure 1, where $\omega_c = 2\pi \times 10^6$ rad/s, $k_f = 2000\pi$, and $k_p = \pi/2$. **(16)**

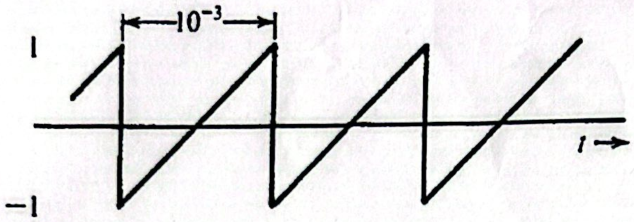

Figure 1. A periodic sawtooth waveform

$A\cos(\omega_c t + k_f \int m(\alpha)d\alpha)$
$A\cos(\omega_c t + k_p m(t))$

$\Delta f = \frac{k_f m_p}{2\pi} = \frac{k_p \dot{m}_p}{2\pi} = \frac{\Delta\omega}{2\pi}$

(i) **Write** equations for $\varphi_{FM}(t)$ and $\varphi_{PM}(t)$.

(ii) **Determine** the frequency deviation Δf.

(iii) **Estimate** the bandwidth of the angle modulated signals, both B_{FM} and B_{PM}. Assume the bandwidth of $m(t)$ to be the fifth harmonic frequency of $m(t)$.

$2(\Delta f + B)$
15k+2

(iv) **Sketch** $\varphi_{FM}(t)$ and $\varphi_{PM}(t)$ for this signal $m(t)$.

(b) Expose the values for an Armstrong indirect FM modulator (for the block diagram in Figure 2) to generate an FM carrier with a carrier frequency of 98.1 MHz and $\Delta f = 75$ kHz. The following equipment is available: **(4)**

(i) NBFM generator with local oscillator of 100 kHz and $\Delta f = 10$ Hz.
(ii) An adjustable frequency oscillator in the range of 10 to 11 MHz.
(iii) A BPF and frequency doublers, triplers, and quintuplers ($\times$ 5).

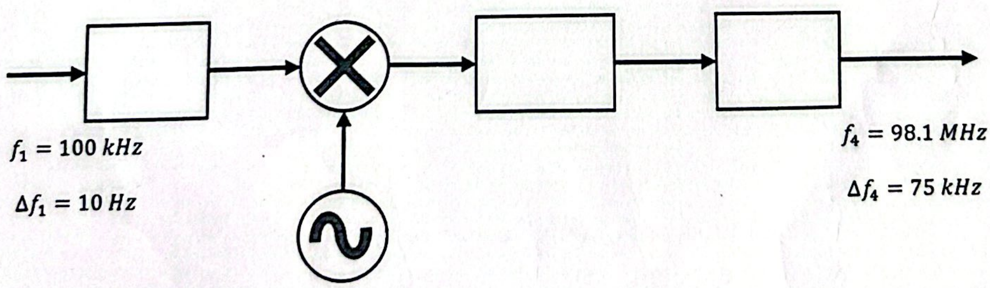

Figure 2. Block Diagram of the Indirect Armstrong FM Modulator

100k + 150
= 18M
7.5M
951k